



The effect of serious game on the level of anxiety, fear, knowledge of hospitalized children and level of anxiety in their parents

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ABSTRACT

Background: Serious games can be used to increase the knowledge level of hospitalized children and reduce their anxiety levels and fear of medical procedures.

Objectives: This randomized controlled study was conducted to examine the effects of the developed serious game on children's anxiety, fear and knowledge levels and their parents' anxiety levels.

Methods: For this randomized controlled trial, a pre-test-post-test experimental study method was used, guided by the CONSORT checklist. The sample consisted of 70 children aged 8–12 years and their parents. Socio-demographic Data Collection Form, Fear Scale of Medical Treatment, Clinical Knowledge Form, State-trait Anxiety Inventory for Children and for Adults were used to data collection.

Results: As a result of the study, it has been found that serious gaming affects the state and trait anxiety levels, medical treatment fear levels and clinical knowledge levels of hospitalized children aged 8–12 years. It was determined that the state anxiety of the parents of the children participating in the study decreased, their trait anxiety is not affected.

Conclusion: In children aged 8–12 years, the use of serious game increased knowledge and decreased anxiety and fear of medical procedures. Larger studies involving parents are needed.

Application to practice: Nurses are healthcare professionals who play a key role in managing anxiety through non-pharmacological interventions. Opportunities should be created for nurses to use artificial intelligence in their own initiatives and transfer it to the clinical environment. It is the first Turkish serious game developed by nurses.

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Introduction

The effects of the hospitalization experience on children vary from one child to another. The reasons for this discrepancy can be attributed to differences in the child's cognitive development, age, duration of illness, perception of the concept of illness, level of physical restrictions, presence of pain, type of illness, anxiety levels in parents, readiness for hospitalization, and varying levels of anxiety experienced during hospitalization. Hospitalization leads to anxiety in all children. The reasons why children of all ages show high levels of anxiety symptoms during hospitalization are inadequate preparation for hospitalization, parents' anxiety level, children's lack of knowledge about clinical procedures (lack of child-centered information), and fear of medical procedures (Delvecchio et al., 2019; Johnson et al., 2021; Salmi & Hanson, 2021).

The parents who have to stay with the child throughout the hospitalization process are involved in the process from the first day. This situation paves the way for the development of anxiety in parents (Kaynak & Özçelik, 2020). As invasive nursing procedures are initiated on children, the anxiety levels in parents begin to rise (Pragholapati et al., 2020). Anxiety is a learned and contagious emotion. Therefore, the anxiety experienced by parents begins to affect the child. It has been shown that as parental anxiety increases in children aged 9–12, so does the anxiety in children (Kaynak & Özçelik, 2020).

An additional factor contributing to heightened anxiety levels in hospitalized children is the inaccuracy or insufficiency of health information. It is well-established that providing accurate information about medical procedures results in more favorable emotional and physical outcomes for children (Bray et al., 2019). As children's knowledge about upcoming procedures increases, their anxiety levels decrease. Children typically acquire health information from their parents or individuals in their immediate environment. When the information provider fails to convey information in a manner suitable for the child's age or if their own knowledge is inaccurate or incomplete, it can

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lead to increased anxiety levels in children. Expectations among children in the 8–12 age group for receiving procedure-related information directly from healthcare professionals are often unmet (Bray et al., 2019; Bray et al., 2022). Therefore, it is essential that health information sessions for children be child-centered and delivered directly by healthcare professionals (Bray et al., 2019).

When the hospitalization experience is not effectively managed in children, it can lead to the development of negative emotions and behaviors such as mistrust of healthcare personnel, refusal to collaborate with healthcare staff, regression, lack of collaboration, withdrawal, aggression, crying, and anxious behaviors (Delvecchio et al., 2019; Li et al., 2016). Health professionals have emphasized the importance of preparing children for clinical procedures through various interventions to reduce their anxiety while highlighting the significance of informing them through a child-centered approach (Bray et al., 2020). When examining interventions aimed at reducing children's fears and anxieties during hospitalization, commonly utilized methods include visual-auditory or solely auditory presentations, videos, development of books-brochures, and drawing pictures (Agüero-Millán et al., 2023; Suleiman-Martos et al., 2022; Tomaszek et al., 2019; Tsao et al., 2017). Although these methods have demonstrated various benefits, it has been noted that computer and web-based interventions are the most effective way to prepare hospitalized children for the hospital environment, with their usage in the healthcare field increasing in recent years (Bray et al., 2020).

Recent studies have incorporated the latest version of web-based interventions, known as serious gaming initiatives. With the increasing interest of children in games and the internet, serious games are recognized as an opportunity to be utilized in anxiety management, which is a nursing intervention (Buffel et al., 2019; Matthyssens et al., 2020). Serious game interventions in child health are observed to focus on children undergoing surgery and those with chronic illnesses (Buffel et al., 2019; Eijlers et al., 2019). Despite the recommendation for the use of serious gaming interventions by healthcare professionals to help hospitalized children adapt to their environment (Jin & Kim, 2019), literature review revealed no serious game specifically designed for this purpose. The aim of the study is to examine the effects of the developed serious game on the anxiety, fear and knowledge of children aged 8–12 who were hospitalized for the first time and the anxiety levels of their parents.

The hypotheses of the study are; H1: The anxiety levels of the children in the study group are lower than the control group, H2: The clinical knowledge level of the children in the study group is higher than the control group, H3: The medical procedure fear level of the children in the study group is lower than the control group, H4: The anxiety level of the parents of children in the study group is lower than the control group.

Methods

This study followed Consolidated Standards of Reporting Trials.

Design

An pretest–posttest experimental study design was used. This study followed Consolidated Standards of Reporting Trials.

Study setting and sampling

The research was conducted in the pediatric service of a university hospital located in a western province of Turkey between October 2022 and March 2023. A power analysis was performed using the G*Power-3.1.9.2 software to determine the required sample size for the study. Taking into account a Type I error of 0.05, Type II error of 0.20 (80 % power), and a medium effect size, a total sample size of 70 children and parents (experimental: 35, control: 35) was calculated.

The data for the study were collected in-person. After providing necessary explanations, the forms were given to the children and parents to fill out, and they were given time to complete them. Once the participants finished filling out the forms, the forms were collected.

In the study, patients were divided into experimental and control groups. Simple randomization technique was used to create the groups. Since patient rooms were for two people, randomization was made according to room number (This method was chosen to prevent the possibility of having both control and experimental group patients in the same room). Every patient who met the sample selection criteria was included in the study. The ward secretary randomly assigned patients to rooms. Patients placed in the experimental room were assigned to the intervention group, and patients placed in the control room were assigned to the control group. The patients and the ward secretary did not know whether the rooms in the ward were experimental or control rooms.

In the study, the lottery method was used to prevent selection bias in the random assignment process. As a result of using the lottery method, it was determined that odd-numbered rooms would constitute the experimental group, and even-numbered rooms would constitute the control group. Randomization was done by another academician who was not involved in the study.

Inclusion/exclusion criteria

The inclusion criteria for the study are as follows: no previous hospitalization, aged between 8 and 12 years, receiving treatment while hospitalized for at least three days, voluntary participation, no visit due to trauma, absence of oncological issues, absence of speech impairments, absence of comprehension difficulties, and absence of cognitive impairments.

Study interventions

The aim of the study is to examine the effects of the developed serious game on the anxiety, fear and knowledge of children aged 8–12 who were hospitalized for the first time and the anxiety levels of their parents.

Development of the serious game

The scenario of the game was crafted by researchers. During the development phase of the game content, insights were sought from three academic faculty members specializing in pediatric nursing. Support for the technology and infrastructure to be utilized in the game was obtained from a professional software company. The title of the game, deemed to be appealing to children, memorable, simple, and fitting for its purpose, was chosen as “Hospital Adventure,” with the protagonist named “Melomet.” To enable gameplay and data storage, a server was rented where the game was deployed.

The purpose of the game is to impart information prepared by researchers to the children playing the game. In the game; introduction of the tools and equipment found in the hospital room (patient wristband, patient file, serum hanger, etc.), tools and equipment frequently used by nurses during treatment (syringe, branule, serum pump, etc.) and small information about the facilities in the hospital (activity room, hospital school, etc.) have been prepared. It was thought that increasing children's knowledge about the hospital environment, the medical supplies used by nurses, and the facilities in the hospital would reduce their anxiety levels. The objective of the child playing the game is to acquire knowledge by playing levels containing different games. At the end of the game, points are earned by answering questions based on the acquired knowledge. With the earned points, each child creates their own avatar at the end of the game. The game can be played on a web page. To ensure game security, entry into the game is possible with a username and password.

The serious game utilized visuals designed based on pediatric ward environments for the web interfaces. The game consists of two stages. The first stage is the gaming section, which includes games across 4 different levels. The second stage of the game is the question section where the player can create an avatar based on the principle of answering questions correctly.

When the player enters the game, the game segment, which is the first stage of the game, begins. In the initial level of the game, the player is required to match puzzles. The theme of this level is based on the room of a child in the hospital. The purpose of this level is to introduce the hospital room and help the player learn the purpose of the items in the hospital room. The player is expected to drag the puzzle pieces at the bottom of the game screen to the main picture. In this level, if the player drags the correct puzzle piece, it matches the main picture and the color of the area in the main picture becomes vivid. When the puzzle piece is dragged to the correct area in the game, a sound is played indicating that the player has made a correct match. If the player tries to make an incorrect match, a sound is played indicating that the player is wrong and the puzzle piece returns to its previous place on the screen. The aim of this level is to attract the player's attention by using visual and auditory interactions. In the game, information pop-ups that are requested to be given are loaded into some puzzle pieces. After the player reads the information requested, he/she closes the pop-up screen by clicking "x" and can continue the game. There are a total of 5 information pop-ups in this level (wheelchair, hospital wristband, hospital, examination table, patient file).

In the second level of the game, the player must find the objects hidden in the interface. The images of 5 hidden objects are given at the bottom of the game interface. In this level, the main image is used in black and white, while the objects to be found are given in color. At the top of the page in the interface, there is the command "Can you find the hidden objects?" The theme of this level is based on the room of a child in the hospital. The aim of this level is to ensure that the player learns the medical equipment used by nurses during treatment. When the hidden objects are found in the main picture and clicked, an information pop-up opens and a sound is played that tells you that you found the correct one. The exact image of the correctly found hidden object is given and the nurse's purpose in the treatment is explained. After the player reads the information requested, he can close the pop-up by clicking "x" and continue the game. The hidden objects found are colored in the main picture. There are a total of 5 hidden objects in this section (serum, syringe, vascular access, pump device, medicine).

In the 3rd level of the game, card matching is required. There are 12 cards in the game interface and 6 visuals (there are two of each visual). At the beginning of the page in the interface, there is the command "Let's find the same pictures". In the theme of this level, cards with a nurse visual showing a question mark are placed on a flat surface. The aim of this level is to ensure that the player learns about the nurse and the medical equipment the nurse uses during follow-up, and to reduce their fears if any. In this level, if the player finds and clicks on the same visual one after the other, a sound is played indicating that they did the right thing and a pop-up with descriptive information about the nurses and the medical equipment they use during follow-up appears. After the player reads the information requested, they close the pop-up by clicking "x" and continue the game. The visual that is clicked correctly in succession remains open in the interface. There are a total of 6 information pop-ups in this section (degree, blood pressure monitor, nurse, doctor, stethoscope, nurse uniform).

The final level of the game features a balloon-popping activity. The participant is expected to pop all the balloons on the game interface. At the top of the page on the interface, there is a command that says "Come on, pop the balloons, get the information!" The theme of this level was created based on the hospital playroom. The aim of this level is to allow the player to discover the fun aspects of staying in hospital. In this level, information pop-ups that are requested to be given inside some balloons are loaded into the game. When the balloon

containing the information pop-up is popped, a sound is played indicating that it is the correct balloon. After the player reads the information to be given, he closes the pop-up by clicking on the "x" and continues the game. There are a total of 4 information pop-ups in this section (playroom, friendship, hospital school, playing games). (Fig. 1).

Upon completion of the first stage of the game, the second stage consisting of questions begins. The second stage of the game consists of questions. In this section, 15 questions are asked that include the information given in the first stage. The questions contain 3 options. The player must answer the question correctly and quickly. If the participant answers the questions correctly, the scoring is done according to the answering time (5 points if they answer in 0–5 s, 4 points if they answer in 5–10 s, 3 points if they answer in 10–15 s, 1 point for 15 and above). Since the player must answer the questions correctly and quickly in order to get a high score in this section, the aim is for the player to remember and learn the information that needs to be given in the game. These features are presented in different categories according to the points. For players who wish to see a wide range of options, it is necessary to answer the question correctly and quickly. It is thought that by motivating the player in this way to answer the question correctly and encourage replay, their motivation will increase. At the end of the questions, the participant can see the avatar they created and the total points earned from the game. They can also view the avatars created by other players and their scores. The opportunity is provided for the player to see their own ranking and the rankings of all players participating in the game. This stage of the game aims to create inter-player competition and enhance the replay value of the game (Fig. 2).

Pilot implementation of the research

The pilot study was conducted with 10 children. Children included in the pilot study meet the study inclusion criteria. The children included in the pilot study were not part of the research sample. Feedback from children in the pilot group was generally positive. One technical issue was reported, such as a question not being fully read in the game. Based on the recommendations of the children, some updates were made to the game, and the final version of the game was released.

Implementation phase of the research initiative

At this stage, the designed serious game was played for 3 days from the hospitalization of the experimental group, and the standard clinical hospitalization process was not intervened in the control group.

Children who had never been hospitalized before (surgery, chronic or acute illness, etc.) were included in the study. The children included in the study were played a serious game designed on their first day of hospitalization. The codes for the game were then given to the child and they were allowed to play for as long and as many times as they wanted for 3 days. During this hospitalization, the children were given various nursing interventions such as opening an intravenous line, taking blood samples, and taking urine tests.

The diagnoses of the children included in the study included urinary tract infection, diarrhea, tooth abscess, sore throat, abdominal pain, etc. The children included in the study were hospitalized in the general pediatrics ward. The general purpose of hospitalization was IV antibiotic treatment. They had not undergone any surgical operation.

When children are admitted to the hospital, one of their parents is asked to stay with them. There is an adult with each child who is hospitalized. A child who is with one of his/her parents at all hours of the day is affected by his/her parent. It is thought that the anxiety level of the child is affected by the anxiety level of the parent.

Data collection tools

In the research, the following data collection tools were utilized: Sociodemographic Data Collection Form, Clinical Knowledge Form,



Fig. 1. Interfaces in phase 1 of the game.

Medical Procedures Fear Scale (MPFS), State-Trait Anxiety Inventory for Children (STAI-C), State-Trait Anxiety Inventory (STAI) for assessing anxiety levels.

Sociodemographic data collection form

The form consists of a total of 14 questions, including information on sociodemographic characteristics of children and parents such as age, gender, and education level. The form was developed by researchers through a review of the literature (Bossavit & Parsons, 2018; Buffel et al., 2019; Ingadottir et al., 2017).

Clinical knowledge form

Researchers developed the form to determine the hospital and clinic-related information of the children participating in the study. During the development of the form, researchers aimed to introduce the tools and equipment that children would encounter most frequently in a hospital setting. To assess the comprehensibility and feasibility of the developed form, opinions were obtained from seven experts. The agreement among experts was examined using the Kendall's W test.

There was no statistically significant difference found between the scores given by the experts to the items on the information form (Kendall's W = 0.120, $p = 0.660$). The form consists of 20 questions about procedures conducted in the clinic, implemented procedures, medical tools used by nurses, and healthcare professionals. The questions in the form are multiple choice. Each question has 3 options. There are one right, one wrong and "I don't know" options.

While creating this form, the items that children routinely see in the hospital room (patient wristband, patient file, IV stand, etc.), the tools and equipment frequently used by nurses in treatment (syringe, branule, IV pump, etc.) and the facilities available in the hospital (activity room, hospital school, etc.) were taken into consideration.

Medical Procedures Fear Scale (MPFS)

The Medical Procedures Fear Scale (MPFS) was developed by Marion Broome and colleagues in 1987. MPFS is designed to measure children's fears related to medical procedures. The scale consists of four subscales: procedural, environmental, personal, and interpersonal, totaling 29 questions. The validity and reliability study of the scale in our country was conducted by Alak (1993). In the reliability analyses of this study,

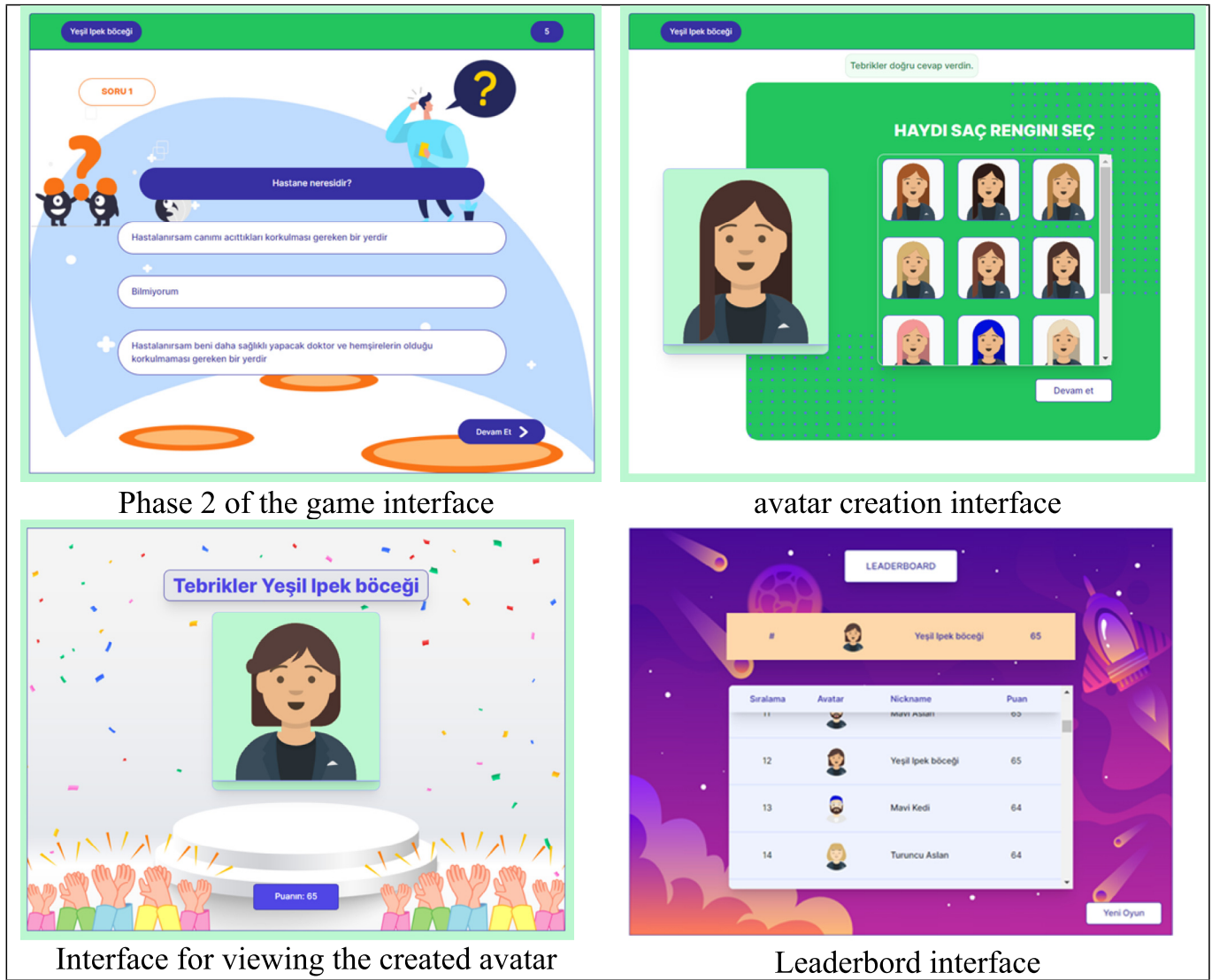


Fig. 2. Interfaces in phase 2 of the game.

the Spearman-Brown reliability coefficient was found to be 0.93. The 29 questions in the scale are in a three-point Likert-type format. The scale consists of twenty-nine questions and is a three-choice Likert type. The child included in the study is asked to choose one of the following statements for each item in the scale: “I am not afraid at all” (1), “I am a little afraid” (2), “I am very afraid” (3). The minimum score that participants can receive in calculating the scale score is 29, while the maximum score is 87. When evaluating the scores, children who score 29 are considered to have no fear of medical procedures, those scoring between 29 and 58 are considered to have some fear, and those scoring between 58 and 87 are considered to have a high level of fear towards medical procedures.

The State-Trait Anxiety Inventory for Children (STAI-C)

The State-Trait Anxiety Inventory for Children (STAI-C) was developed by Spielberg (1973). Validity and reliability studies of the Turkish version have been conducted. The STAI-C can be administered to children aged 9–12 years. The inventory consists of two parts. The State Anxiety Inventory for Children (SAI-CH) aims to measure how a child “generally” feels anxious. This section of the scale contains 20 items with questions in a three-point Likert scale format. There are reverse-scored items

in this section as well. An increase in scores from the scale indicates an increase in the child’s continuous feeling of anxiety.

When the scale is applied, the child is asked to answer with one of the following options: (1) “almost never”, (2) “sometimes” and (3) “often”, indicating the frequency of the feelings or behaviors in the items in the scale. When the scale is scored, 3 points are given to the items for which the expression “often” is selected, while 1 point is given to the items for which the expression “almost never” is selected. A minimum of 20 points and a maximum of 60 points are obtained from the scale. An increase in the scores obtained from the scale indicates an increase in the anxiety that the child constantly feels. The Cronbach’s alpha coefficient was found to be 0.81 in the Turkish version of the Trait Anxiety Inventory for Children (Özusta, 1995).

The Trait Anxiety Inventory for Children (TAI—C) comprises 20 items in this section, with questions rated on a three-point Likert scale. The TAI-C is administered to assess how children are feeling at the “current” time of measurement. The inventory includes items that are reverse-scored. An increase in scores from the inventory signifies an elevation in anxiety levels experienced by the child at the time of assessment.

When applying the (TAI—C), the child is asked to mark the most appropriate statement for him/herself from the statements such as “I feel very calm, I feel calm, I do not feel calm” on the scale. There are direct

and reversed items on the scale. Direct statements express negative emotions, while reversed items express positive emotions. When scoring the negative emotions felt, 3 points are given if they are felt a lot, and 1 point if they are not felt at all. A minimum of 20 points and a maximum of 60 points are obtained from the scale. The increase in the scores obtained from the scale indicates the increase in anxiety in the child at the time the scale is applied. The Cronbach's alpha coefficient in the Turkish version of the Trait Anxiety Inventory for Children was found to be 0.82 (Özusta, 1995).

The State-Trait Anxiety Inventory (STAI): (adult)

The STAI is used to assess individuals' anxiety levels in two subdimensions, instantaneously and continuously. The scale was developed by Spielberger (1985) and has been adapted into Turkish. The internal consistency and reliability of the scales were found to be between 0.83 and 0.92 for the State Anxiety Scale and 0.86–0.92 for the Trait Anxiety Scale. There is no cutoff point for scale scores.

The State Anxiety Scale: The State Anxiety Scale describes the anxiety the individual feels “at the moment” the scale is administered. It is a 4-point Likert-type scale consisting of twenty items. When the State Anxiety Scale is used, individuals are asked to evaluate how they feel “at the moment.” The scale includes items that are reverse-scored. As the individual's score on the scale increases, it is understood that the anxiety level at the time of measurement is also increasing.

The Trait Anxiety Scale: The Trait Anxiety Scale aims to measure how individuals “generally” feel in terms of anxiety propensity. It is a 4-point Likert-type scale consisting of twenty items. When the Trait Anxiety Scale is administered, individuals are asked to evaluate how they “generally” feel about themselves. As the individual's score on the scale increases, it is understood that there is an increase in the continuous level of anxiety.

When scoring reversed statements, items with a weight value of 1 are given 4 points, and items with a weight value of 4 are given 1 point. If the total score obtained from the scales is between 0 and 19, it indicates that the individual does not feel anxiety, if it is between 20 and 39, it indicates that the individual feels mild anxiety, if it is between 40 and 59, it indicates that the individual feels moderate anxiety, and if it is between 60 and 79, it indicates that the individual feels severe anxiety.

Timeline of events

The Sociodemographic Data Collection Form, Clinical Knowledge Form, MPFS, and STAI-C were administered to the children in the experimental group, while the STAI was administered to the parents of the children in the experimental group on the day of hospitalization. When patients who were recommended to be hospitalized came to their rooms in the pediatric ward (before invasive procedures), the forms used for pretesting were filled out. Following the completion of the forms, a serious game developed by the researchers was played with the children. Afterwards, hospitalization procedures and nursing interventions began (opening vascular access, taking anamnesis, etc.). The children were provided with a link to the game and were granted access to it for three days. The game can be played on both computers and tablets. On the third day of hospitalization, the Clinical Knowledge Form, MPFS, and STAI-C were administered to the children, while the parents were again given the STAI questionnaire.

The Sociodemographic Data Collection Form, Clinical Knowledge Form, MPFS, and STAI-C were administered to the children in the control group, while the STAI was administered to the parents of the children in the control group on the day of hospitalization. No intervention was made in the standard clinical hospitalization process for the control group. Forms were applied to the control group. No games were played. Clinical procedural procedures and nursing interventions were applied (anamnesis, opening vascular access, blood

collection, etc.). Forms were applied again on the 3rd day of hospitalization. On the third day of hospitalization for the children in the control group, the Clinical Knowledge Form, MPFS, and STAI-C were administered, and the parents were again given the STAI questionnaire. The CONSORT (Schulz et al., 2010) diagram is shown in Fig. 3.

Statistical analysis

The data obtained in the study were analyzed using the free trial version of SPSS Statistics (Statistical Package for Social Sciences) for Windows 25.0 program. Descriptive statistical methods (number, percentage, minimum, maximum, median, mean, standard deviation) were employed in the data evaluation process. To observe the differences in qualitative data among groups, Chi-square analysis was conducted during the data analysis.

The normal distribution of the data was assessed using the Shapiro-Wilk test. Independent samples *t*-test was conducted for data showing a normal distribution, while the Mann-Whitney *U* test was employed for non-normally distributed data. To assess whether there was a significant difference between two dependent groups, paired samples *t*-test was utilized for normally distributed data, and the Wilcoxon test was applied for data that did not exhibit normal distribution. A significance level of $p < 0.05$ was adopted.

Ethical considerations

Ethical approval for the study was obtained from the ethical board for non-interventional research of Pamukkale University (Approval date: June 8, 2021; Approval number: E-60116787-020-61,005). Necessary permissions were also obtained from the institution where the research was conducted. For the scales used in the research, permission was obtained from the relevant authors via e-mail before the study. Consent was obtained from all children who agreed to participate verbally and from their parents in writing. All procedures were conducted in compliance with the Helsinki Declaration. The study was registered on [ClinicalTrials.gov](https://clinicaltrials.gov/ct2/show/study/NCT05775640) (NCT05775640).

Results

Socio-demographic characteristics of the sample

The questions in Table 1 (except the last two questions) were asked to the parent. The last two questions were asked to the child.

It was determined that the majority of children in the intervention group are 8 (25.7 %) to 9 (25.7 %) years old. 40.0 % of the participants have some knowledge about the procedures to be performed in the hospital, while 11.4 % have received training on hospitalization. The socio-demographic characteristics of the patients are detailed in Table 1. Analysis results indicate that children in the experimental and control groups are homogenous in terms of socio-demographic characteristics ($p > 0.05$) (Table 1).

Results corresponding to the children

In the research, no significant difference was found between the groups in the pre-test ($t = -0.352$; $p = 0.726$) and post-test ($t = -1.421$; $p = 0.160$) mean scores for continuous anxiety ($p > 0.05$). However, a significant difference within the experimental group was observed in the pre-test and post-test mean scores for continuous anxiety scale ($t = 2.434$; $p:0.020$). Following the intervention, a decrease in the mean scores of continuous anxiety was observed in the experimental group.

While no significant difference was found in the pre-test mean scores for state anxiety between the experimental and control groups ($t = 0.491$; $p = 0.625$), a significant difference was observed in the post-test mean scores ($t = -2.341$; $p = 0.022$) ($p < 0.05$). Following

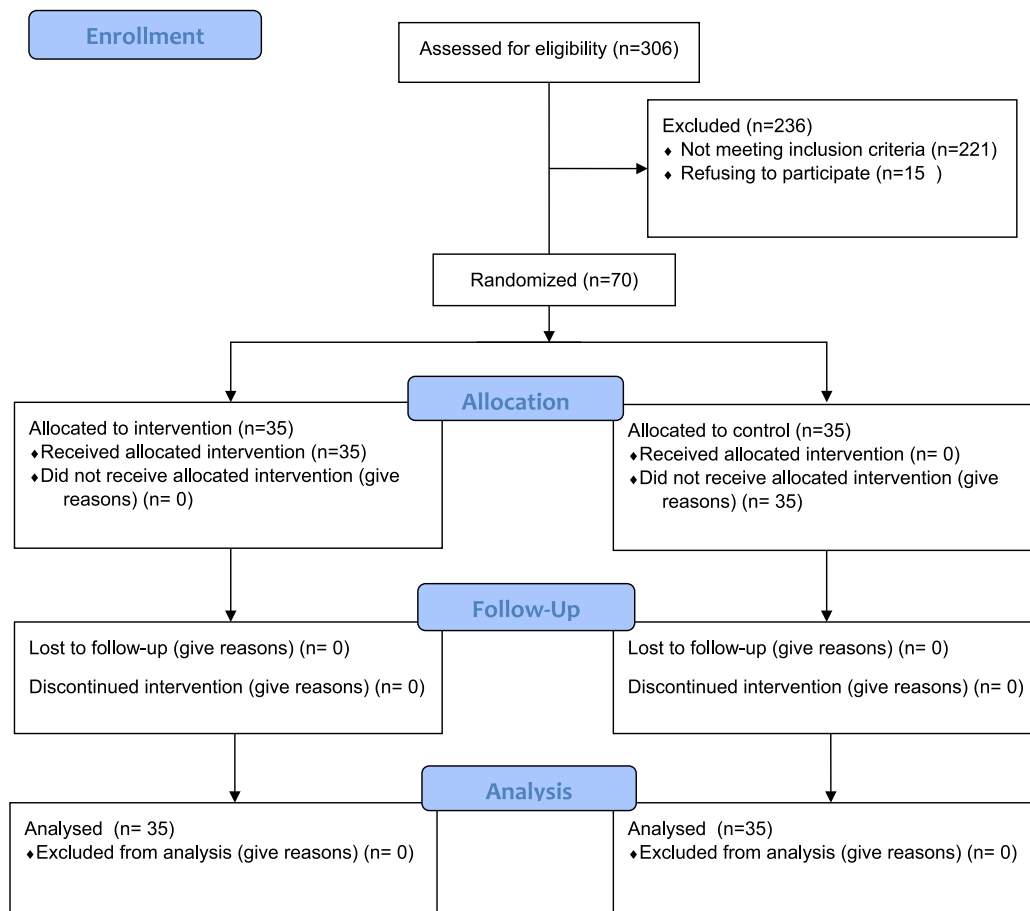


Fig. 3. CONSORT Diagram.

the intervention, a decrease in the mean scores of state anxiety was observed in the experimental group (Table 2).

In the research, it was determined that there was no statistically significant difference in the pre-test ($Z = -0.247$; $p = 0.805$) and post-test ($Z = -1.771$; $p = 0.077$) mean scores for MPFS between the experimental and control groups. Within-group analysis showed that while there was no statistically significant difference in the post-test mean scores for MPFS in the control group ($Z = -1.617$; $p = 0.106$), a statistically significant difference was found in the post-test mean scores for MPFS in the experimental group following the intervention ($Z = -4.347$; $p = 0.000$) ($p < 0.05$). A decrease in the mean scores for MPFS was observed in the experimental group after the intervention (Table 3).

In the environmental fear subscale scores of the MPFS scale, no significant difference was observed between the groups in the pre-test ($Z = -0.983$; $p = 0.326$), but in the post-test ($Z = -2.094$; $p = 0.036$), it was noted that the mean scores of the experimental group were lower. A decrease in the mean scores for environmental fear was observed in the experimental group following the intervention. There was no statistically significant difference between the groups in the scale's other subscales ($p > 0.05$), and the distribution is detailed in Table 3.

In the study, the responses given by the participants to the questions in the clinical knowledge form were evaluated. It was observed that in the experimental group, the rates of knowing question 1 ($p = 0.000$), question 2 ($p = 0.002$), question 3 ($p = 0.000$), question 4 ($p = 0.002$), question 5 ($p = 0.000$), question 6 ($p = 0.031$), question 7 ($p = 0.000$), question 8 ($p = 0.001$), question 10 ($p = 0.039$), question 13 ($p = 0.031$), question 14 ($p = 0.016$), question 15 ($p = 0.031$), question 16 ($p = 0.004$), question 18 ($p = 0.000$), and question 19 ($p =$

0.000) increased after the intervention (Table 4). In the control group, only the rates of knowing question 18 and question 19 increased, while there was no statistically significant difference in other questions ($p > 0.05$).

Results corresponding to the parents

Statistically significant differences between the average scores of continuous anxiety of parents in the experimental and control groups were not observed in the pre-test ($t = -1.218$; $p = 0.227$) and post-test ($t = -1.589$; $p = 0.117$) in both groups ($p > 0.05$).

When examining the differences between the average state anxiety scores of parents in the experimental and control groups; while no significant difference was found in the pre-test scores of the experimental and control groups ($p = 0.233$), a statistically significant difference was observed in the post-test scores ($p = 0.007$) following the intervention applied to the experimental group ($p < 0.05$). It was observed that the average state anxiety scores of parents after the intervention in the experimental group were lower compared to the control group (Table 5).

Discussion

Children of all ages admitted to the hospital are known to exhibit high levels of anxiety symptoms (Koukourikos et al., 2015). Considering the impacts of anxiety on children's psychological and physiological parameters, all efforts aimed at reducing anxiety become critically important.

In our study, it was observed that following the serious game intervention administered to the children in the experimental group, their

Table 1
Distribution of children and parents in the intervention and control groups according to descriptive characteristics.

Variables		intervention (n = 35)		Control (n = 35)		Statistical Analysis*
		n	%	n	%	
Age	8	9	25,7	7	20,0	$\chi^2 = 1583$ $p = 0,850$
	9	9	25,7	7	20,0	
	10	6	17,1	8	22,9	
	11	6	17,1	9	25,7	
	12	5	14,4	4	11,4	
Sex	Female	18	51,4	18	51,4	$\chi^2 = 0,000$ $p = 1,000$
	Male	17	48,6	17	48,6	
person who answered the form	Mother	32	91,4	28	80,0	$\chi^2 = 1867$ $p = 0,306$
	Father	3	8,6	7	20,0	
	Respiratory system disease	9	25,7	13	37,1	
Reason for hospitalization	Digestive system diseases	13	37,1	10	28,6	$\chi^2 = 1683$ $p = 0,844$
	Urinary tract disease	4	11,4	3	8,6	
	Musculoskeletal diseases	7	20,0	6	17,1	
	Other	2	5,7	3	8,6	
Does your child play digital games?	Yes	30	85,7	34	97,1	$\chi^2 = 2917$ $p = 0,198$
	No	5	14,3	1	2,9	
Economic status	Income less than expenses	15	42,9	15	42,9	$\chi^2 = 0,461$ $p = 1,000$
	Income equal to expenses	19	54,3	18	51,4	
	Income more than expenditure	1	2,9	2	5,7	
Mother's education	Primary school	17	48,6	14	40,0	$\chi^2 = 0,884$ $p = 0,824$
	Secondary school	6	17,1	8	22,9	
	High school	10	28,6	10	28,6	
	University and above	2	5,7	3	8,6	
Father's education	Primary school	14	40,0	9	25,7	$\chi^2 = 4752$ $p = 0,201$
	Secondary school	11	31,4	8	22,9	
	High school	6	17,1	14	40,0	
	University and above	4	11,4	4	11,4	
Mother's occupation	Worker	5	14,3	5	14,3	$\chi^2 = 1057$ $p = 0,731$
	Housewife/retired	29	82,9	27	77,1	
	Officer	1	2,9	3	8,6	
Father's occupation	Officer	5	14,3	6	17,1	$\chi^2 = 1791$ $p = 0,863$
	Worker	21	60,0	17	48,6	
	Self-employed	6	17,1	10	28,6	
	Not working/retired	3	8,6	2	5,7	
Number of siblings	1	3	8,6	5	14,3	$\chi^2 = 3102$ $p = 0,209$
	2	16	45,7	21	60,0	
	3 and above	16	45,7	9	25,7	
Number of children	1	8	22,9	11	31,4	$\chi^2 = 1166$ $p = 0,590$
	2	19	54,3	19	54,3	
	3 and above	8	22,9	5	14,3	
**Was there any explanation about the illness, the hospital etc. During this hospitalization?	Yes	4	11,4	4	11,4	$\chi^2 = 0,000$ $p = 1,000$
	No	31	88,6	31	88,6	
**Do you have any information about what will happen to you in the hospital?	I don't know anything	10	28,6	8	22,9	$\chi^2 = 1039$ $p = 0,858$
	I have some information	14	40,0	18	51,4	
	I have information	7	20,0	6	17,1	
	I know everything	4	11,4	3	8,6	

* $p < 0,05$; χ^2 : Kikare.

** Questions asked to children.

levels of persistent anxiety decreased. Within the experimental group, it was found that the levels of persistent anxiety were lower in intra-group comparisons. Conversely, in the control group, there was a minimal increase in levels of persistent anxiety. When examining another parameter of our study, state anxiety levels, it was noticed that after the serious game intervention provided to the children in the experimental group, their state anxiety levels were found to be low in both intra-group and inter-group comparisons. On the other hand, there was a minimal increase in state anxiety levels among children in the control group. Reviewing studies involving the use of serious game interventions to reduce anxiety in hospitalized children in the literature, it appears that these interventions are mainly focused on children with chronic conditions such as diabetes, obesity, autism spectrum disorder, asthma (Carlier et al., 2020; Holtz et al., 2018), or children scheduled for surgery (Buffel et al., 2019; Ingadottir et al., 2022; Matthysens et al., 2020). It has been suggested that serious game interventions in

child healthcare should be utilized for all hospitalized children by healthcare professionals (Jin & Kim, 2019). However, only one study has been found in the literature addressing this topic. The serious game-based platform developed by Bray and colleagues, named "Xploro DTx" (2020), was implemented on 80 children aged 8 to 14 undergoing planned clinical procedures in the UK. Significant reduction in anxiety levels was observed post-intervention, which aligns with the findings of our study. Nevertheless, in our study, we specifically sampled children hospitalized for the first time, considering the potential impact of hospitalization frequency on anxiety levels. It is suggested that serious game interventions could be employed to reduce anxiety in children aged 8–12 admitted to the hospital for the first time.

Despite being influenced by various factors, medical procedure fear in children is thought to be mitigated through the collaborative efforts of the child, nurse, and family, given that nurses are typically the first point of contact for the child in the clinic, responsible for admitting

Table 2

Mean trait and state anxiety scores of children in the intervention and control groups.

Variables		Intervention (n = 35)			Control (n = 35)			Test** and p
		Median (min-max)	$\bar{x}$	SS	Median (min-max)	$\bar{x}$	SS	
Trait Anxiety	Pretest	35 (22–49)	34,57	6,99	35 (24–47)	35,14	6,58	t = −0,352 p = 0,726
	Posttest	35 (21–47)	33,69	6,75	37 (23–48)	35,94	6,53	t = −1421 p = 0,160
	Test Values*** p	t = 2434 0,020*			t = −1413 0,167			
State anxiety	Pretest	46 (37–57)	46,11	6,87	48 (39–53)	46,62	6,78	t = 0,491 p = 0,625
	Posttest	45 (38–53)	44,74	4,33	47 (38–52)	46,94	5,26	t = −2341 p = 0,022*
	Test Values*** p	t = 2124 0,041*			t = ,756 0,455			

* p < 0,05.

** t: independent sample t.

*** t: dependent sample t.

them to the ward, and carrying out invasive/non-invasive procedures. According to our study findings, it was observed that after playing the serious game developed for the experimental group, levels of medical procedure fear decreased in children. While digital-based interventions are highlighted in the literature for reducing medical procedure fear and anxiety in children (Farrier et al., 2020; Kılıç et al., 2021), only one study has been found that specifically explores the impact of serious game interventions on medical procedure fear in children (Taddio et al., 2022).

It is known that providing accurate information about medical procedures leads to more positive emotional and physical outcomes in children (Bray et al., 2019). Based on this knowledge, our study adopted a

child-centered approach in the serious game we developed. The game was designed for the child to actively access the information by playing it themselves. As a result of our study, it was observed that children in the experimental group showed an increase in clinical knowledge levels after playing the serious game. There have been no studies found in the literature investigating the impact of serious games on the knowledge levels of children hospitalized for the first time. Current studies show the development of serious games for children undergoing surgery or hospitalized for chronic conditions (Mack et al., 2020; Sarasmita et al., 2021; Vrancken, De Gryse, & Ve Spooren, 2021; Ingadottir et al., 2022). Due to ongoing research in this field, serious games have been

Table 3

Comparison MPFS of total and subscale mean scores of children in the intervention and control groups.

Variables		Intervention (n = 35)			Control (n = 35)			Test** and p
		Median (min-max)	$\bar{x}$	SS	Median (min-max)	$\bar{x}$	SS	
Procedural	Pretest	12 (9–20)	12,49	2,89	12 (9–19)	12,43	2,74	Z = −0,118 p = 0,906
	Posttest	12 (9–17)	11,60	2,17	12 (9–21)	12,80	3,09	Z = −1488 p = 0,137
	Test Values*** p	Z = −3758 0,000*			Z = −0,907 0,365			
Environmental	Pretest	10 (7–20)	10,31	3,31	10 (7–21)	10,97	3,43	Z = −0,983 p = 0,326
	Posttest	9 (7–19)	9,74	3,02	10 (8–20)	10,91	3,12	Z = −2094 p = 0,036*
	Test Values*** p	Z = −3002 0,003*			Z = −0,413 0,679			
Intrapersonal	Pretest	5 (4–10)	5,74	1,65	5 (4–11)	6,03	2,24	Z = −0,078 p = 0,937
	Posttest	5 (4–10)	5,60	1,85	6 (4–12)	6,09	2,06	Z = −1081 p = 0,280
	Test Values*** p	Z = −1500 0,134			Z = −0,369 0,712			
Interpersonal	Pretest	12 (9–21)	13,11	3,32	12 (9–24)	13,09	3,80	Z = −0,296 p = 0,767
	Posttest	10 (9–21)	11,77	3,14	11 (9–27)	13,20	4,46	Z = −1066 p = 0,286
	Test Values*** p	Z = −4086 0,000*			Z = −0,203 0,839			
CMFS Total	Pretest	40 (29–68)	41,66	9,87	41 (29–71)	42,51	10,73	Z = −0,247 p = 0,805
	Posttest	36 (29–64)	38,71	8,71	41 (31–77)	43,00	10,96	Z = −1771 p = 0,077
	Test Values*** p	Z = −4347 0,000*			Z = −1617 0,106			

* p < 0,05.

** Z: Mann-Whitney.

*** Z: Wilcoxon.

Table 4

Comparison of clinical knowledge status of children in the intervention and control groups.

	Pretest	Posttest					
		Intervention (n = 35)			Control (n = 35)		
		False	True	p **	False	True	p **
Question 1	False	0	12	0,000*	8	1	1000
	True	0	23		0	26	
Question 2	False	0	10	0,002*	7	0	1000
	True	0	25		1	27	
Question 3	False	0	12	0,000*	14	4	0,125
	True	0	23		0	17	
Question 4	False	1	13	0,002*	10	5	1000
	True	1	20		4	16	
Question 5	False	2	17	0,000*	13	3	0,625
	True	1	15		1	18	
Question 6	False	1	6	0,031*	10	3	0,625
	True	0	28		1	21	
Question 7	False	1	16	0,000*	15	3	1000
	True	0	18		2	15	
Question 8	False	0	11	0,001*	15	2	1000
	True	0	24		3	15	
Question 9	False	1	4	0,125	2	1	1000
	True	0	30		2	30	
Question 10	False	14	8	0,039*	31	1	1000
	True	1	12		0	3	
Question 11	False	0	5	0,063	1	0	1000
	True	0	30		0	34	
Question 12	False	0	4	0,125	1	0	1000
	True	0	31		0	34	
Question 13	False	1	6	0,031*	9	1	1000
	True	0	28		1	24	
Question 14	False	2	7	0,016*	7	2	1000
	True	0	26		2	24	
Question 15	False	3	6	0,031*	20	2	1000
	True	0	26		2	11	
Question 16	False	2	9	0,004*	2	1	0,625
	True	0	24		3	29	
Question 17	False	0	7	0,070	2	1	1000
	True	1	27		1	31	
Question 18	False	2	12	0,000*	4	8	0,039*
	True	0	21		1	22	
Question 19	False	1	17	0,000*	0	23	0,000*
	True	0	17		1	11	
Question 20	False	0	5	0,063	1	4	0,125
	True	0	30		0	30	

* p < 0,05.

** Mc Nemar.

developed for children in recent years, but their clinical trials are not yet completed (Ingadottir et al., 2022; Sarasmita et al., 2021). In studies examining various age groups of children who played serious games, high clinical knowledge scores were found (Mack et al., 2020; Sarasmita

et al., 2021), and in a qualitative study conducted by Koukourikos et al. (2015) with children aged 8–12 presenting to the hospital, children expressed their desire to have information about procedures when they come to the hospital using the “write and tell” technique. Despite no such studies found in the literature, it is believed that our study will contribute to filling this gap in this area.

Anxiety is a learned and also a contagious emotion, thereby children are influenced by the anxiety levels of parents (Kaynak & Özçelik, 2020). Due to the relationship between parental anxiety and children's anxiety, parents are also an important target audience for non-pharmacological interventions frequently used by nurses (Verschuere et al., 2019). Furthermore, it is considered a parameter that needs to be addressed in nursing research that parents' anxiety levels increase most during nursing interventions applied to children (Pragholapati et al., 2020). According to the findings of our study, the levels of continuous anxiety in parents of children in the experimental and control groups are similar before and after the intervention. It has been understood that the serious game intervention applied to children does not affect the continuous anxiety levels of the parents present with them. When the situational anxiety levels of parents of children in our study as another parameter were examined, a decrease in situational anxiety levels was observed in the parents of children who engaged in serious game after the intervention.

When examining the current literature regarding the anxiety of parents who stay with hospitalized children, studies primarily focus on determining the anxiety levels of parents and investigating the factors causing anxiety. In the literature, studies aiming to reduce the anxiety of parents of hospitalized children have explored the impact of social support systems (Boyden et al., 2020; Gise & Cohen, 2022), parental involvement in the care of the sick child (Çamur et al., 2021), and addressing the educational and counseling needs of parents (Zarei & Negarandeh, 2021). These interventions have been shown to reduce parents' anxiety; however, there is a lack of research on the impact of serious game interventions on parental anxiety. The study findings contribute to the literature in this field.

Strength and limitations of the work

In constructing the sample for the study, limitations include the reliance on a single center, the assumption of the accuracy of responses provided by children and parents to scale questions, and the inability to play the game on mobile phones.

Strengths of the research; strengths of the research; The research is a randomized controlled experimental study, the developed serious game “Hospital Adventure” is the first Turkish game developed for children in the 8–12 age group who are hospitalized for the first time, the study aims to provide child-centered information, The developed serious

Table 5

Distribution of trait and state anxiety scores of parents of children in the intervention and control groups.

		Intervention (n = 35)			Control (n = 35)			Test** and p
		Medyan (min-max)	$\bar{x}$	SS	Medyan (min-max)	$\bar{x}$	SS	
Trait Anxiety	Pretest	44 (26-67)	44,94	9,17	49 (31-69)	47,43	7,85	$t = -1218$
	Posttest	46 (26-61)	45,06	8,58	47 (35-68)	48,06	7,16	$p = 0,227$
	Test Values*** p	$t = 0,187$ 0,853			$t = -1334$ 0,191			$t = -1589$ $p = 0,117$
State Anxiety	Pretest	44 (21-59)	41,49	10,71	47 (24-59)	44,37	9,08	$Z = -1193$
	Posttest	41 (21-58)	39,89	9,31	47 (29-57)	45,46	7,37	$p = 0,233$
	Test Values*** p	$t = 1775$ 0,085			$Z = -1333$ 0,183			$t = -2776$ p = 0,007*

* p < 0,05.

** Z: Mann-Whitney; t: independent sample t.

*** Z: Wilcoxon; t: dependent sample t.

game includes many rich visual and audio elements such as background, pictures and sound recordings to attract the child's attention and is a Web-based application.

Implications for clinical practice

Games developed for children need to be integrated into clinics and environments where children can play need to be created. Pediatric nurses are key in determining children's needs, so they need to be supported in serious game design.

It is recommended to develop games targeting different age groups and diseases and to develop serious games in different languages. Studies involving families are limited. Large studies involving families are needed (Gilboa-Negari, Abu-Kaf, Huss, & Hain G. ve Moser, 2017; Pragholaipati et al., 2020; Zarei & Negarandeh, 2021).

Conclusions

The results of this study revealed that, children who played serious games had higher levels of clinical knowledge, lower trait/state anxiety, and less fear of medical procedures. Additionally, it was shown that parents' state anxiety levels decreased.

The serious game designed in this study appears to benefit children aged 8 to 12 years who are hospitalized for the first time and their parents. Serious games can be an effective educational method and can draw children's attention since they are entertaining, safe, and simple to employ for hospitalized children. Therefore, it is recommended to provide the necessary environment for the implementation of serious game methods in children's inpatient wards. Also, the impact of the serious game on hospitalized children in various settings should be further investigated in future studies.

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Data analiz

All data analysis of the study was carried out by an expert statistical company.

CRedit authorship contribution statement

Hilal Parlak Sert: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Hatice Başkale:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

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